Curriculum Vitae of Xin Cui

Ph.D. in Geophysics

Laboratory for Seismology and Physics of Earth's interior
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Research Interests

- Machine Learning in Seismology: ML applications to event clustering and new seismological discoveries.
- Earthquake Physics: Evolution and properties of foreshocks, aftershocks, and mainshocks.
- Earthquake Cycle: Simulation of coseismic and postseismic deformation.

Education

- 2026/2-current, Postdoctoral Researcher in Geophysics, USTC, Hefei, China
  Advisor: Prof. Zefeng Li
- 2020/9-2025/12, Ph.D. Candidate in Geophysics, USTC, Hefei, China
  Advisor: Prof. Zefeng Li
- 2024/9-2025/7, Visiting Student, Université Côte d'Azur, Geoazur Laboratory, Nice, France
  Advisor: Prof. Jean-Paul Ampuero
- 2019/6-2019/9, Visiting Student, California Institute of Technology, Pasadena, USA
  Advisor: Prof. Robert Clayton
- 2016/9-2020/6, B.S. in Geophysics USTC, Hefei, China
  Advisor: Prof. Zefeng Li and Prof. Yan Hu

Awards & Honors

2025	President's Special Award, Chinese Academy of Sciences, China
2025	Dean's Award, School of Earth and Space Sciences, University of Science and Technology of China, China
2023	Outstanding Student Presentation Award, 2023 Annual Meeting of American Geophysical Union (AGU), USA
2023	Best Student Presentation Award, 2023 Annual Meeting of AI for Seismology Conference, Hefei, China
2023	Best Poster Award, 2023 Annual Meeting of International Professionals for the Advancement of Chinese Earth Sciences (IPACES), Hefei, China
2023	National Scholarship Graduate, USTC, China
2022	National Scholarship Graduate, USTC, China

2020–2023	Outstanding Student Scholarships, USTC, China (Four times)
2020	Graduate with honor: USTC, China

Received Funds

- Youth Student Basic Research Project (PhD Student), National Natural Science Foundation of China, 300,000 Chinese Yuan, PI, 2024/01-2025/12
- The 2024 China Association for Science and Technology (CAST) Young Talent Support Program - PhD Student Special Plan, 2024/01-2025/12

Publications

*corresponding author

8. Hu, Y., **Cui, X.** & Li, Z. (2026). Mainshock-induced stress changes modulate initial aftershocks on complex branching faults of the 2019 Ridgecrest earthquake. *Geophysical Research Letters*, *53*, e2026GL122144. doi:10.1029/2026GL122144
7. Yang, S., Hu, Y., Zhao, B., Sang, C., Liu, Y., **Cui, X.**, Chen, Y. & Liu, Z. (2026). Terrestrial water trend changes in the North China Plain and Shaanxi-Gansu regions based on GNSS vertical data. *Chinese Journal of Geophysics*, *69*(3), 1005-1017. doi:10.6038/cjg2025S0710
6. Sang, C., Hu, Y., Yang, S., Wang, K. & **Cui, X.** (2025). Coseismic slip of the 2024 Mw 7.3 Hualien earthquake constrained by GNSS observations and its implications for fault interactions. *Seismological Research Letters*, *97*(2A), 977-985. doi:10.1785/0220250013
5. **Cui, X.** Li, Z.*, Ampuero, J. & Louis, D. (2025). Does foreshock identification depend on seismic monitoring capability? *Geophysical Research Letters*, *52*, e2025GL115394. doi:10.1029/2025GL115394
4. Liu, Y. **Cui, X.** Hu, Y.*, Zhang, J. & Chen, Y. (2024). Integrated investigation on heterogeneous lower crust rheology in Kyushu and afterslip behavior following the 2016 Mw7.1 Kumamoto earthquake. *Geophysical Research Letters*, *51*, e2023GL107606. doi:10.1029/2023GL107606
3. **Cui, X.** Hu, Y. Ma, S. Li, Z.*, Liu, G. & Huang, H. (2024). Bridging supervised and unsupervised learning to build volcano-seismicity classifiers in Kilauea, Hawaii. *Seismological Research Letters*, *95*(3), 1849-1857 doi:10.1785/0220230251
2. **Cui, X.** Li, Z.*& Hu, Y. (2023). Similar seismic moment release process for shallow and deep earthquakes. *Nature Geoscience*, *16*, 454-460 doi:10.1038/s41561-023-01176-5
1. **Cui, X.** Li, Z.*& Huang, H. (2021). Subdivision of seismicity beneath the summit region of Kilauea volcano: Implications for the preparation process of the 2018 eruption. *Geophysical Research Letters*, *48*, e2021GL094698. doi:10.1029/2021GL094698

Manuscripts in Preparation

*corresponding author

2. **Cui, X.** Li, Z.*Ampuero, J. & Louis, D.Understanding Your Catalogs: Automated Earthquake Sequence Extraction, Classification and Characterization, in preparation.

1. **Cui, X.** Li, Z.*& Ampuero, J.Moho depth controls earthquake stress drop in Southern California, in preparation.

Meeting Abstracts

5. **Cui, X.** Li, Z. Ampuero, J. & Louis, D. (2025). Influence of Seismic Monitoring Capability on Foreshock Identification in California: A Comparative Analysis of Methods. 2025 EGU Meeting, Vienna, Austria. No. EGU25-10735.
4. **Cui, X.** Li, Z. & Ma, S. (2024). Moho depth controls earthquake stress drop in Southern California 2024 IPACES Annual Meeting, Beijing, China.
3. **Cui, X.** & Li, Z. (2023). On the Physical Mechanism of Foreshock Sequences in South California. 2023 AGU Fall Meeting, San Fransisco, CA, USA. ID: S31A-05.
2. **Cui, X.** & Li, Z. (2023). Exploring the Predictability of Fault Seismicity with Machine Learning. 2023 IPACES Annual Meeting, Hefei, China.
1. **Cui, X.** & Li, Z. (2021). Are shallow, intermediate-depth, deep-focus EQs distinguishable from source time functions? 2021 Annual Meeting of Asia Oceania Geosciences Society, Online.

Field Experience

- 2021/07/28–2021/08/10, Field Observation Internship in the Tibetan Plateau, China

Teaching and service

- Teaching assistant, Undergraduate Mechanics Course (2019)
- Organizer, Geophysics Student Seminar at USTC (2020-2022)
- Organizer of Geophysics Section, USTC's Science and Technology Week (2022-2024)